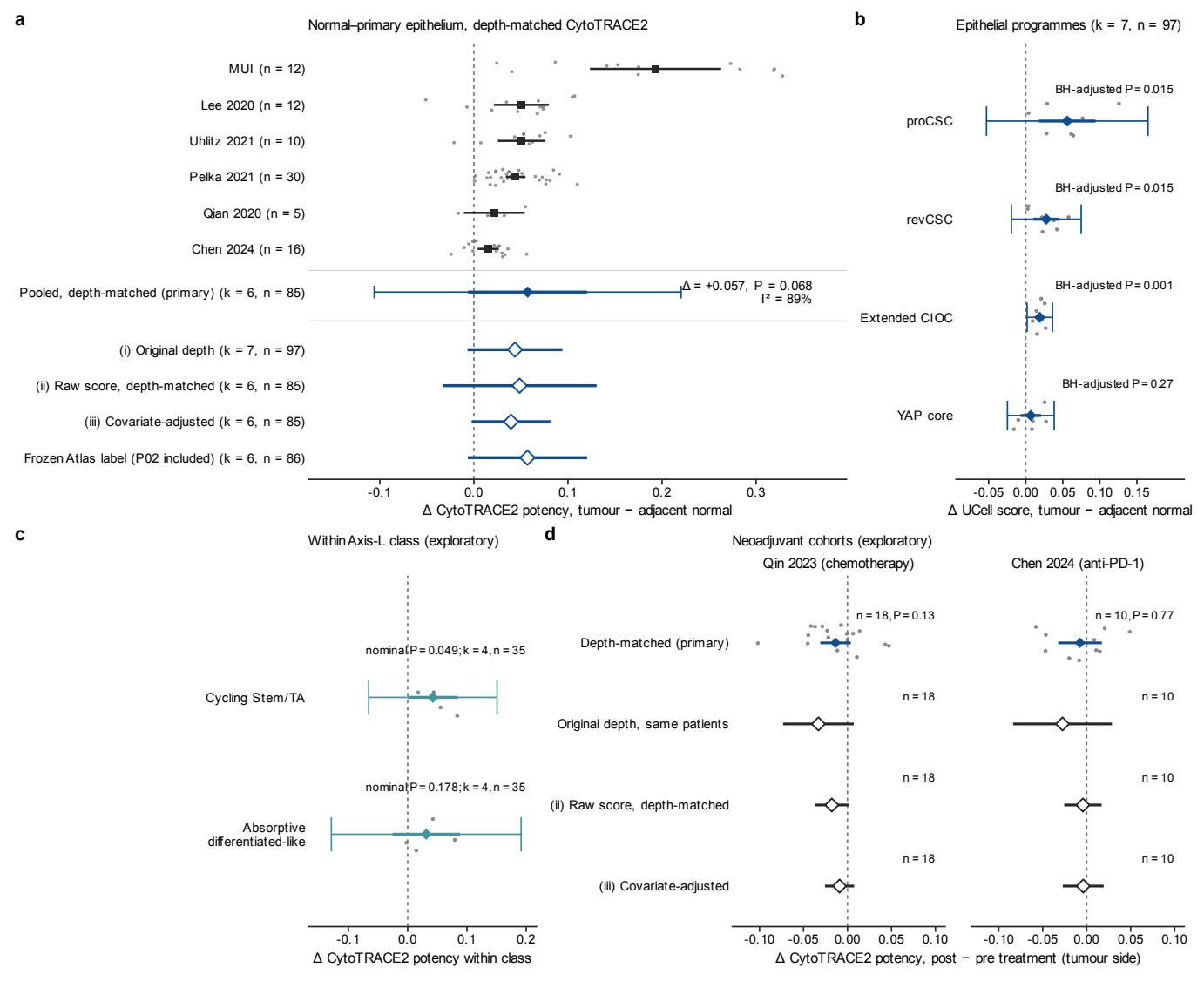
a
Normal–primary epithelium, depth-matched CytoTRACE2
MUI (n = 12)
Lee 2020 (n = 12)
Uhlitz 2021 (n = 10)
Pelka 2021 (n = 30)
Qian 2020 (n = 5)
Chen 2024 (n = 16)
Pooled, depth-matched (primary) (k = 6, n = 85)
Δ = +0.057, P = 0.068
I² = 89%
(i) Original depth (k = 7, n = 97)
(ii) Raw score, depth-matched (k = 6, n = 85)
(iii) Covariate-adjusted (k = 6, n = 85)
Frozen Atlas label (P02 included) (k = 6, n = 86)
-0.1
0.0
0.1
0.2
0.3
Δ CytoTRACE2 potency, tumour − adjacent normal
b
Epithelial programmes (k = 7, n = 97)
BH-adjusted P = 0.015
proCSC
BH-adjusted P = 0.015
revCSC
BH-adjusted P = 0.001
Extended CIOC
BH-adjusted P = 0.27
YAP core
-0.05
0.00
0.05
0.10
0.15
Δ UCell score, tumour − adjacent normal
c
Within Axis-L class (exploratory)
nominal P = 0.049; k = 4, n = 35
Cycling Stem/TA
nominal P = 0.178; k = 4, n = 35
Absorptive differentiated-like
-0.1
0.0
0.1
0.2
Δ CytoTRACE2 potency within class
d
Neoadjuvant cohorts (exploratory)
Qin 2023 (chemotherapy)
Chen 2024 (anti-PD-1)
n = 18, P = 0.13
Depth-matched (primary)
n = 10, P = 0.77
n = 18
Original depth, same patients
n = 10
n = 18
(ii) Raw score, depth-matched
n = 10
n = 18
(iii) Covariate-adjusted
n = 10
-0.10
-0.05
0.00
0.05
0.10
Δ CytoTRACE2 potency, post − pre treatment (tumour side)